

PokeVault Arbitrage & AI Pre-Grade Predetermination Manual

Master Operating Specification, Multi-Marketplace Landed Cost Engine, 5-Second Real-Time Polling Buffer, and AI Pixel-Defect Computer Vision Inspection

Target System: PokeVault Web Engine	Production URL: pokedex-hub-lime.vercel.app	Revision: v3.4.0 Quantum Core
Council Auth: Geek Squad NetNavi	Arbitrage Loop: 5000ms Reactive Polling	Status: Production Active (No Beta Wall)

1. Executive Operational Overview

The PokeVault Platform is a multi-agent, high-throughput arbitrage ecosystem engineered to identify, verify, and execute the lowest-cost acquisitions for Pokémon TCG singles and slabs across global secondary marketplaces. Operating in tandem with the Geek Squad NetNavi Council, the platform continuously monitors 15+ live liquidity pools, calculates real-time landed costs (accounting for sales tax, estimated shipping, and buyer protection fees), provides an interactive pixel-level AI pre-grade predetermination engine, and delivers 1-click Quick Strike purchasing unencumbered by beta walls.

2. Geek Squad NetNavi Council & Autonomous Roles

Navi Unit	Codename / Focus	Operational Function & Guardrails
Consoul.EXE	Executive Arbiter	Orchestrates consensus across all sub-agents. Governs 4-node quorum for market executions and price dispute resolution.
MerchantNavi.EXE	Marketplace Harvester	Scrapes, parses, and normalizes live listings across 15+ marketplaces. Computes dynamic FX conversion and landed costs.
Bass.EXE	Quick Strike Executioner	High-velocity trade routing engine. Intercepts target buy orders, validates checkout endpoints, and marks sold listings instantaneously.
Apogee.EXE	AI Pre-Grade Visionary	Applies edge, corner, centering (50/50 - 60/40), and surface micro-defect computer vision scans to project graded slab ROI.
Archivum.EXE	Ledger & RAG Scribe	Maintains historical card sales data, pricing trends, pop reports (PSA/BGS/CGC), and training weights in the local knowledge graph.
ProtoMan.EXE	Sentinel & Security Gate	Blocks rogue listings, detects shill bids, flags counterfeit certificates, and guarantees all outbound links resolve to verified vendors.

3. 15+ Live Marketplace Aggregation & Landed Cost Formulation

Monitored Liquidity Pools: TCGplayer (Market / Low / Direct), eBay (Buy-It-Now & Ending Soonest), Cardmarket (EU Low / Trend with live EUR/USD FX conversion), PriceCharting, Troll & Toad, Mercari US, TCG Republic, Yahoo Japan Auctions / Buyee, CardHobby, Beckett Marketplace, Slab-Auction Houses (PWCC/Fanatics, Goldin, Heritage), and Verified LGS inventory feeds.

Mathematical Landed Cost Model:

- Landed_Cost = (Base_Price × FX_Rate) + Est_Shipping + (Base_Price × 0.07 [Est Tax]) + Buyer_Protection_Fee
- All listings are ranked strictly by Landed_Cost ascending. Verified TCGplayer Direct and eBay Top-Rated Plus receive algorithmic priority ranking.

4. Raw Quality vs Professional Slab Multiplier Matrix

The PokeVault grading engine standardizes condition ratings across both ungraded raw cards and certified third-party encapsulation slabs. The \$19.99 submission cost basis and grading fees are factored directly into arbitrage ROI models.

Category	Grade / Condition Tier	Code	Price Mult.	Typical Pop / Flaw Characteristics
Ungraded	Gem-Mint (Raw)	raw_mint	1.10x	Pack-fresh, 55/45 centering, zero visible whitening or holo scratches.
Ungraded	Near Mint (NM)	raw_nm	1.00x	Standard baseline. Minor edge touch or micro-whitening dot on back corner.
Ungraded	Lightly Played (LP)	raw_lp	0.78x	Minor edge wear, subtle surface scuff, or mild foil clouding.
Ungraded	Moderately Played (MP)	raw_mp	0.55x	Moderate border wear, minor creasing or binder dent.
Ungraded	Heavily Played (HP)	raw_hp	0.35x	Significant whitening, surface peeling, or creasing.
Ungraded	Damaged (DMG)	raw_dmg	0.20x	Major bends, water damage, pinholes, or heavy ink wear.
Graded Slab	BGS 10 Black Label	bgs10_black	18.0x+	Quad 10 subgrades (Centering, Surface, Edges, Corners). Ultra premium.
Graded Slab	CGC 10 / PSA 10 Gem Mint	psa10	4.50x - 8.5x	Vintage Era multiplier scales up to 8.5x for WotC Holos / Gold Stars.
Graded Slab	PSA 9 / BGS 9.5 Mint	psa9	1.85x	Near perfect presentation with microscopic flaw.
Graded Slab	PSA 8 / CGC 8.5 NM-MT	psa8	1.30x	High liquidity mid-tier slab. Breakeven point on modern singles.
Graded Slab	PSA 7 Near Mint	psa7	1.05x	Minor vintage premium; negative ROI on modern singles after \$19.99 fee.

5. AI Pre-Grade Predetermination & Pixel Defect Scanner

Operators can click any card visual in PokeVault to initiate Apogee.EXE's AI Pre-Grade Predetermination Engine. The visual scanner performs sub-millimeter computer vision inspection over four primary grading axes:

Inspection Axis	Diagnostic Criteria	Tolerance Threshold for Gem Mint 10
Centering Reticle	Front Left/Right & Top/Bottom border pixel width ratios.	Front: 55/45 to 50/50. Back: 75/25 or better (PSA standard).
Corner Geometry	Die-cut radius fidelity, edge burrs, whitening pins.	0 whitening fibers detected at 400x digital magnification.
Edge Cleanliness	Silvering detection on vintage holos, chipping on modern borders.	Zero rough cuts or foil peeling along top and bottom perimeters.
Surface & Foil	Print lines, roller marks, holo scratches, indentation depth.	Zero indentation pits; holo clarity score ≥ 9.8 / 10.

6. 5-Second Real-Time Polling Buffer & Rapid Sold Inventory Rotation

- **Proactive Listing Pool Prefetching:** The system maintains a warm queue of up to 64 live listings per card query.
- **Rapid Sold Inventory Marking:** Upon clicking 'Buy', Bass.EXE calls markListingSold(item.url). The listing is instantly removed from the active queue, and the UI immediately rotates to display the next cheapest listing without a page reload.
- **Zero Sign-Up Beta Wall:** Direct 1-Click checkout routes immediately to the verified seller listing, eliminating onboarding friction during live trade execution.
- **Universal Link Guarantee:** If an exact SKU URL is unpopulated, the system dynamically constructs direct live market search endpoints across TCGplayer, eBay BIN, and Cardmarket, ensuring 100% of cards have working links.

7. Geek Squad Field Operator Execution SOP

Step	Action	Verification Standard
1	Card Ingestion & Search	Enter Pokémon name, card number, or set. Verify all pricing variants render.
2	Condition Selection	Toggle between Ungraded (Mint to DMG) and Graded (PSA 10 to PSA 7, BGS, CGC).
3	AI Visual Defect Scan	Click card art to open Pre-Grade Modal. Review centering reticle and defect pins.
4	Arbitrage Spread Check	Confirm: Estimated Slab Value - Raw Price - \$19.99 Fee > +\$25.00 Net Spread.
5	Quick Strike Execution	Click 'Buy Cheapest Listing'. Verify direct vendor redirect and inventory auto-advance.

AUTHORIZATION & SIGN-OFF:

Certified by Geek Squad Quantum DevOps & Antigravity Master Control.
All systems operational in production at <https://pokedex-hub-lime.vercel.app/>.